

Lean port sheets

Rev F0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | D-044 lean ports A (PV) and B (battery) + insulation monitor, BUS- referenced

01 Supplies and commands

Live 24 V / 5 V / 3.3 V, BUS- tie, commands

File: sheets/01_lean_io.kicad_sch

02 Port A power path (pv)

Varistor Y + monitor

File: sheets/02_leanA_power.kicad_sch

03 Port A sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/03_leanA_ctrl.kicad_sch

04 Port A coil drives

Contactor: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/04_leanA_drive.kicad_sch

05 Port B power path (battery)

Varistor Y + monitor, aR fuses

File: sheets/05_leanB_power.kicad_sch

06 Port B sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/06_leanB_ctrl.kicad_sch

07 Port B coil drives

Contactor + precharge relay: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/07_leanB_drive.kicad_sch

08 Insulation monitor

PE divider, two switched 992 k strings through CA-IS3417WT

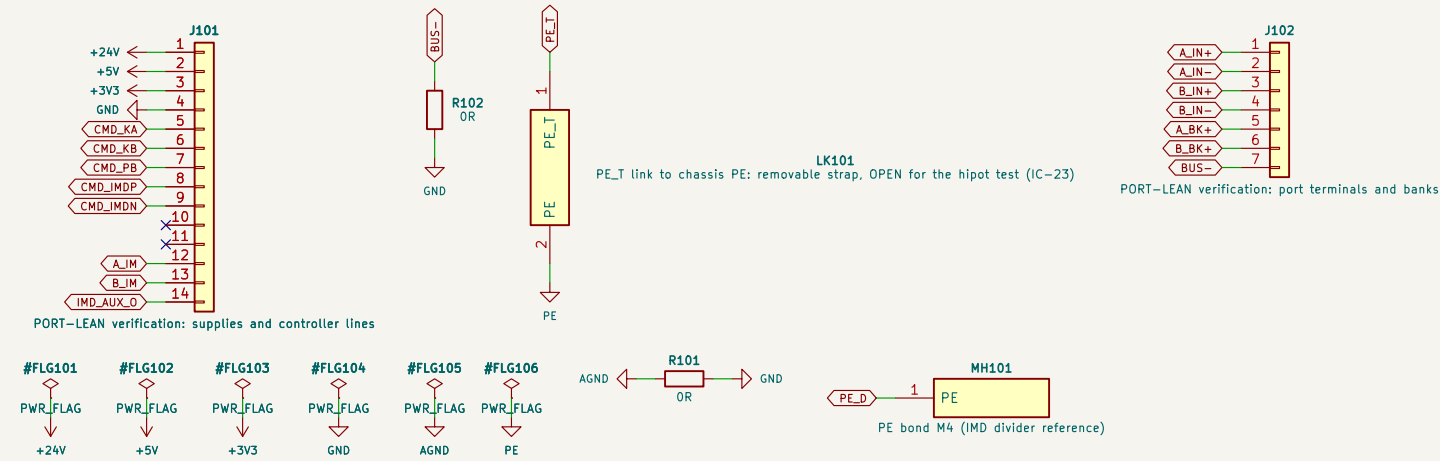
File: sheets/08_lean_imd.kicad_sch

Verification build of the reusable lean-port functions of gen/port.py (lean_port, varistor_y, lean_contactor, lean_precharge, lean_bleeder, lean_dividers, lean_shunt, lean_interlock, lean_imd). The product sheets live on the power board.

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: / File: PORT-LEAN-HOLD.kicad_sch		
Title: Lean port sheets – Cover / Hierarchy		
Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 1/9

SUPPLIES AND COMMANDS



On the power board these come from the AUX-T1 live winding and the controller.

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 2
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /01 Supplies and commands/
File: 01_lean_io.kicad_sch

Title: Lean port sheets – Supplies and commands

Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 2/9

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{LN}+ / A_{LN}-, bank A_{BK}+, No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above [port_spec lean]. Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN+} / A_{IN-}, bank A_{BK+}. No fuse, no precharge: array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

[illegible]

TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up,eff 3793 V pole-PE and pole-pole at in 3 kA (calc., ≤ 4.0 kV $\rightarrow$ reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 k (L/R 2 ms). Monitor loop through both fuses and all three links $\rightarrow$ CNY65B $\rightarrow$ RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open $\rightarrow$ CTRL inhibits/stops.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 4

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /03 Port A sensing and interlocks/

File: 03_leanA_ctrl.kicad_sch

Title: Lean port sheets – Port A sensing and interlocks

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 4/9

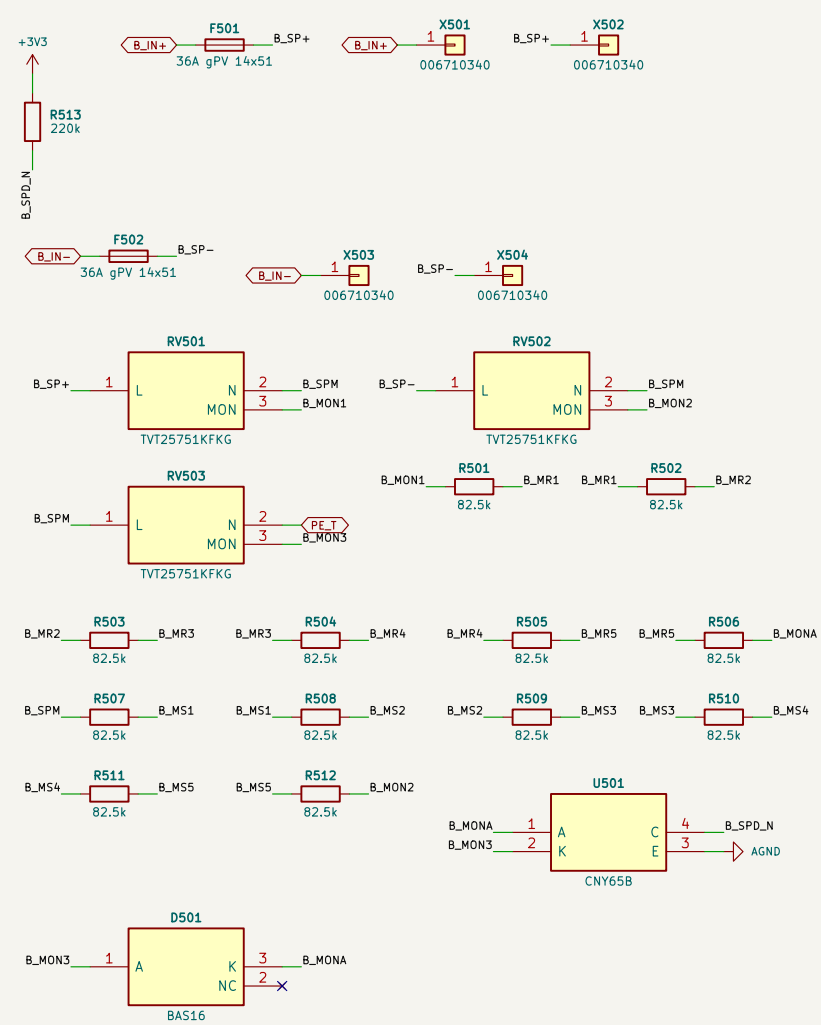
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'		
Sheet 5		
Schematics authored by Chinmoy Bhuyan		
Live side referenced to BUS-; basic HV-PE; functional inside the live side		
Sheet: /04 Port A coil drives/ File: 04_leanA_drive.kicad_sch		
Title: Lean port sheets – Port A coil drives		
Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6	Id: 5/9	

PORT B (BATTERY SOURCE), D-044 LEAN PORT

Terminal side B_{JN+} / B_{JN-}, bank B_{BK+}, HPIE501 250 A aR in both poles (breaks 1.25–50 kA only).
Circuit breaker opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

Diagram illustrating the on-board surge protection circuit for Varistor Y (IC-01, IC-12). The circuit includes a varistor (RV501) connected to the +V5V supply and ground. The varistor is labeled with a 36A gPV 14x51 specification. The circuit also includes a fuse (F501) and a diode (D501) connected to the +V5V supply. The varistor is connected to the ground plane through a 36A gPV 14x51 specification. The circuit is labeled with various components and their values, including RV501, RV502, RV503, RV504, RV505, RV506, RV507, RV508, RV509, RV510, RV511, RV512, RV513, RV514, RV515, RV516, RV517, RV518, RV519, RV520, RV521, RV522, RV523, RV524, RV525, RV526, RV527, RV528, RV529, RV530, RV531, RV532, RV533, RV534, RV535, RV536, RV537, RV538, RV539, RV540, RV541, RV542, RV543, RV544, RV545, RV546, RV547, RV548, RV549, RV550, RV551, RV552, RV553, RV554, RV555, RV556, RV557, RV558, RV559, RV560, RV561, RV562, RV563, RV564, RV565, RV566, RV567, RV568, RV569, RV570, RV571, RV572, RV573, RV574, RV575, RV576, RV577, RV578, RV579, RV580, RV581, RV582, RV583, RV584, RV585, RV586, RV587, RV588, RV589, RV590, RV591, RV592, RV593, RV594, RV595, RV596, RV597, RV598, RV599, RV600, RV601, RV602, RV603, RV604, RV605, RV606, RV607, RV608, RV609, RV610, RV611, RV612, RV613, RV614, RV615, RV616, RV617, RV618, RV619, RV620, RV621, RV622, RV623, RV624, RV625, RV626, RV627, RV628, RV629, RV630, RV631, RV632, 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TV125751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE_T, Up. eff. 3793 V pole-PE and pole-pole at in 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA, Branch

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 7
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /06 Port B sensing and interlocks/
File: 06_leanB_ctrl.kicad_sch

Title: Lean port sheets – Port B sensing and interlocks

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 7/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 8
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /07 Port B coil drives/
File: 07_leanB_drive.kicad_sch

Title: Lean port sheets – Port B coil drives

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 8/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 9

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /08 Insulation monitor/

File: 08_lean_imd.kicad_sch

Title: Lean port sheets – Insulation monitor

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 9/9